****

==========

a▷: X◁F

a▷: X-< F

a▷: X◁F

aj▷: X → Fj → X for each j ∈ S

aj▷ = ai▷ ◦ Fh

◁a: F▷X

◁a: F >- X

◁aj: Fj → X for each j ∈ J

◁aj = Fh ◦ ◁ai for any h: i → j ∈ J

==========

==== Cone ====

Cone

Let F: J → A

Cone

◁a: X◁F

◁aj: X → Fj for each j ∈ J

◁aj = ◁ai ◦ Fh for any h: j → i ∈ J

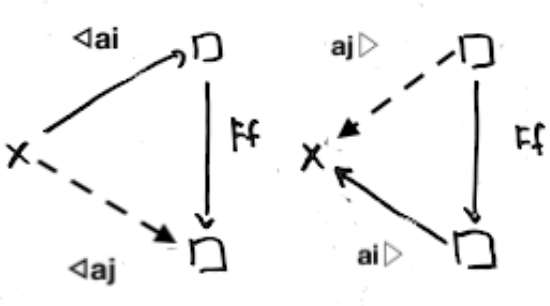
Co-cone

a▷: F▷X

aj▷: Fj → X for each j ∈ J

aj▷ = Fh ◦ ai▷ for any h: i → j ∈ J

Cone Co-cone



Composition

◁a ◦ f: X'◁F for f: X'→X

σ ◦ ◁a: X◁G for σ: F ⇒ G

f ◦ a▷: F▷X' for f: X → X'

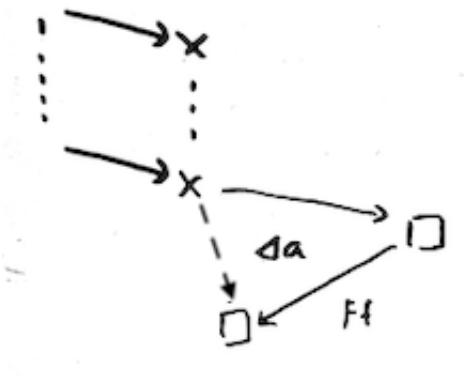
a▷ ◦ σ: G▷X for σ: G ⇒ F

Base cone map

◁a* ∈ [_→X]◁F with base cone ◁F

◁a*: {_→X} → {_◁F}
◁a* (h) = ◁a ◦ h

Base co-cone map
*a▷ ∈ F▷[X→_] with base co-cone F▷
*a▷: {X→_} → {F▷_}
*a▷ (h) = h ◦ a▷



Proposition: natural transformation
Cone is equivalent to

Cone		**Natural transformation**
◁a: X◁F	<=>	α: F ⇒ **Δ**X
Base cone map		**Base map lattice**
*◁a ∈ [_→X] ◁F	<=>	*α ∈ [_→X]⇒F
		*αj ∈ [_→X]→Fj natural in j
		i.e. *αj = *Fh ◦ *αi
		αj = Fh ◦ αi

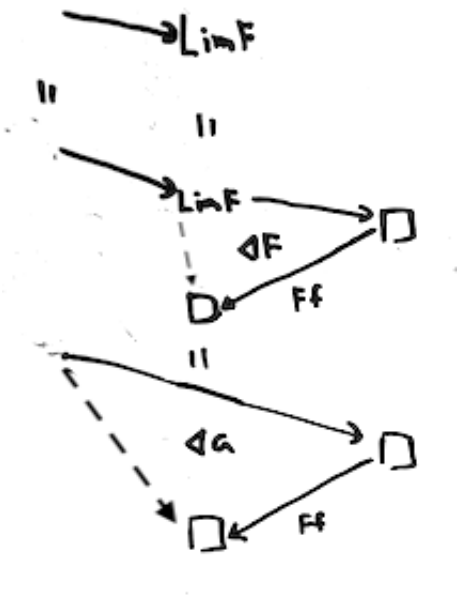
Limit (upper bound)
Colimit (lower bound)
If *◁F is an **isomorphic** co-base map, then the base point is a **limit**:

*◁F ∈ [_→Lim F] ◁F limit

If F▷* is an **isomorphic** base map, then the base point is a **colimit**:

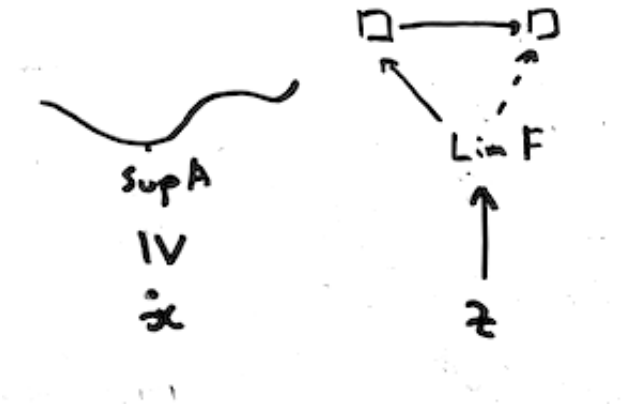
F▷* ∈ F▷[Colim F→_] colimit

Limit



Note:

Limit ~ **sup**
i.e. least upper bound
Colim ~ **inf**
i.e. greatest lower bound

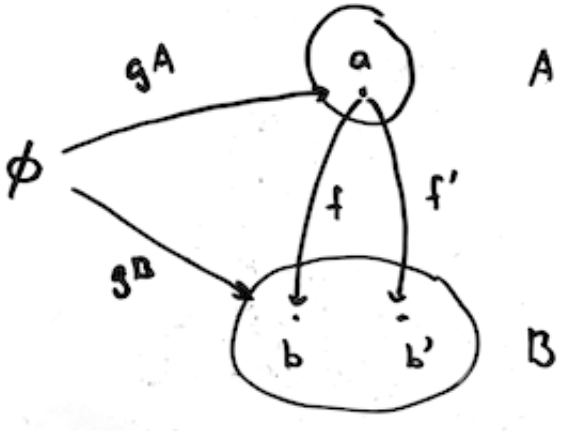


Trivial cone
Trivial limit/colimit
A limit is **trivial** if it is a **initial element**.
Analogously, a colimit is **trivial** if it is a **terminal element**.
In both case, there is only one **trivial cone**.

e.g.
Let
A = { a }
B = { b, b' }
J = {↓ | ↓} parallel arrows

with functor

F = { f | f' }
f, f': A → B
f(a) = b f'(a) = b'



If $\lhd g: X \lhd F$ is a cone, then

$gA: X \to A \qquad \lhd gB: X \to B$
$gB = f \circ gA = f' \circ gA$

But if $x \in X$

$gB(x) = f(gA(x)) = f(a) = b$
$gB(x) = f'(gA(x)) = f'(a) = b'$

=> impossible, so

$X = \varnothing$
$\lhd g: \varnothing \lhd F$ i.e. no map

Theorem: Limit functor
Suppose Lim F exists for all $F: J \to A$
Then limit is a functor

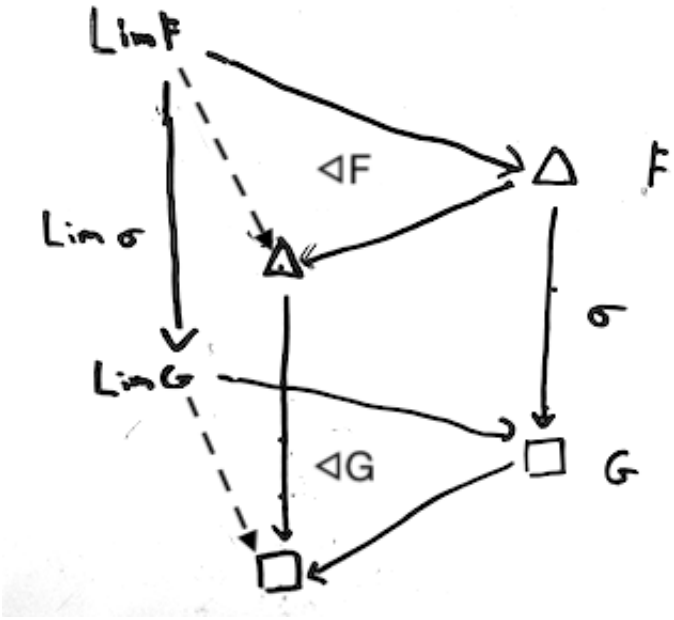
Lim: A^J→A
Lim F object map

Lim σ: Lim F → Lim G arrow map
where $\sigma: F \Rightarrow G$ and unique Lim σ

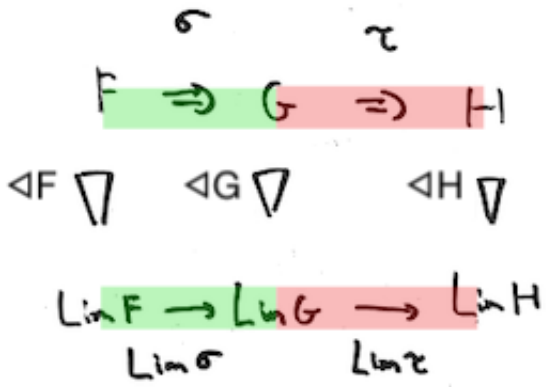
$\sigma \circ \lhd \mathbf{F} = \lhd \mathbf{G} \circ \mathbf{Lim}\ \sigma$

Also, for $X \in A$

Lim X = Lim $\mathbf{\Delta} X$
Lim ($\lhd a$) = Lim ($a: \mathbf{\Delta} X \Rightarrow F$) : Lim X → Lim F
where $\lhd a: X \lhd F$



Composition



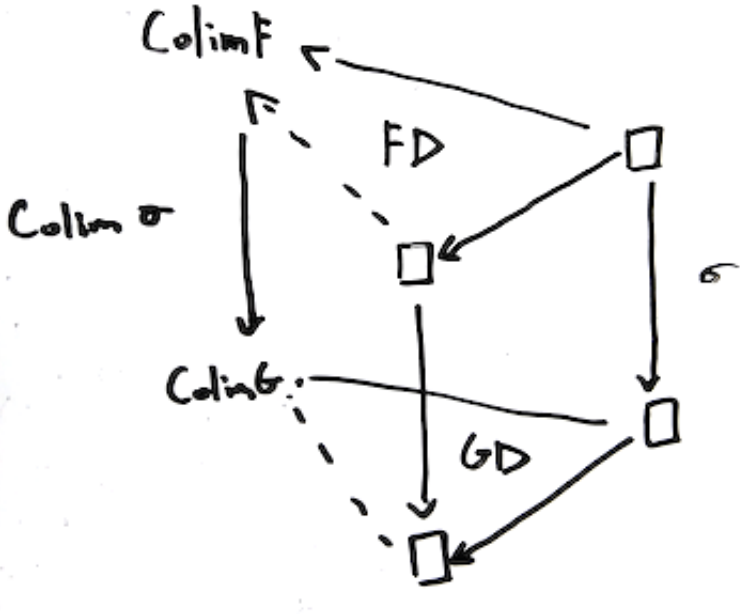
Theorem: Colimit functor
Suppose Colim F exists for all F: J → A
Then

Colim F: A^J → A

Colim σ: Colim F ⇒ Colim G
where σ: F ⇒ G
G▷ ∘ σ = Colim σ ∘ F▷

Let

Colim X = Colim **Δ**X
Colim (a▷: F▷X) = Colim (a: F ⇒ **Δ**X): Colim F → Clim X



==========

Proof

- Lim σ existence

Suppose σ: F ⇒ G. Then

$\sigma \circ \lhd F$: Lim F ⊲ G

=> $\sigma \circ \lhd F = \lhd G \circ \sigma'$ for a unique σ': Lim F→Lim G by Lim G

=> Lim σ = σ'

- composition: Lim τ σ = Lim τ ∘ Lim σ

Let σ: F ⇒ G, τ: G ⇒ H. Then

$\tau \circ \sigma \circ \lhd F = \tau \circ (\lhd G \circ \text{Lim } \sigma)$ by Lim σ

$= (\lhd H \circ \text{Lim } \tau) \circ \text{Lim } \sigma$ by Lim τ

$= \lhd H \circ (\text{Lim } \tau \circ \text{Lim } \sigma)$

=> Lim ($\tau \circ \sigma$) = Lim τ ∘ Lim σ by uniqueness of Lim ($\tau \circ \sigma$)

QED

==========

Theorem: P-family limit

Limit of a family of functors is also a functor.

Let

F: J x P→ A family of functors

Fp = F(·,p): J → A

- **limit functor**: If each Fp has a limit, then the family of limits is a functor:

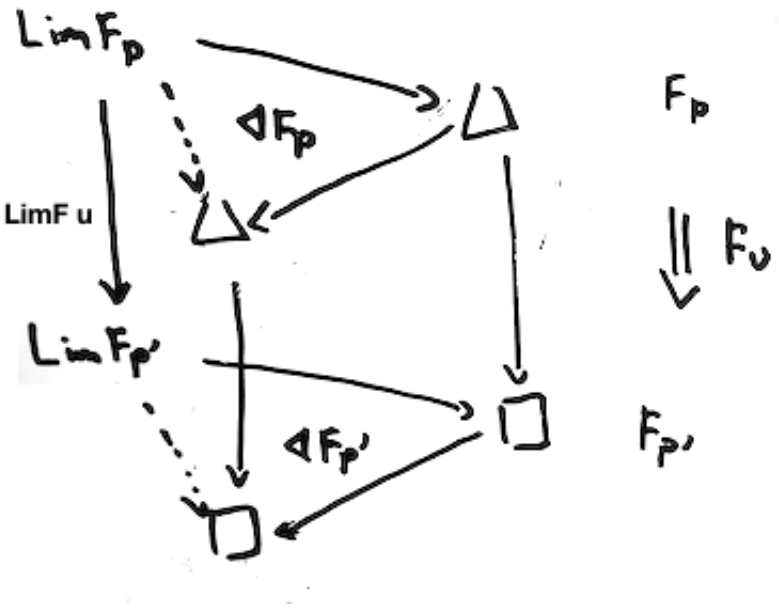
LimF: P → A

LimFp = Lim Fp object map for p ∈ P

LimFu: Lim Fp → Lim Fp' arrow map for u: p → p' ∈ P

such that Fu: Fp ⇒ Fp'

$\lhd Fp' \circ \text{LimF } u = Fu \circ \lhd Fp$



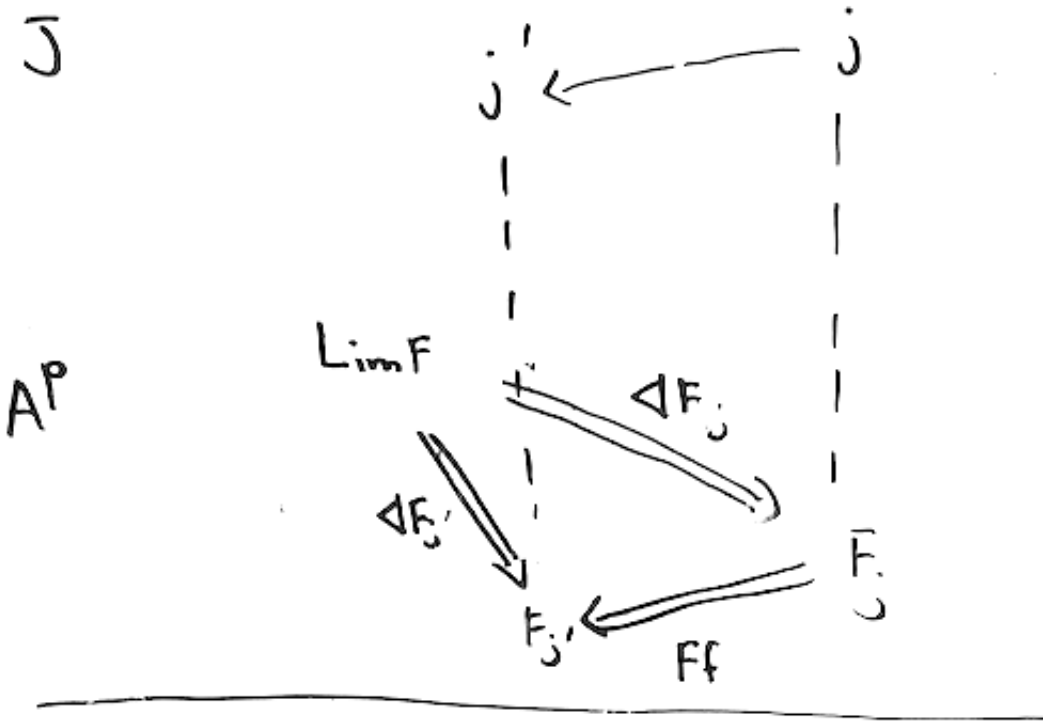
- **limit**: the limit functor itself is a limit of the P-family.
i.e. in the functor category A^P, LimP is a limit of F: J → A^P

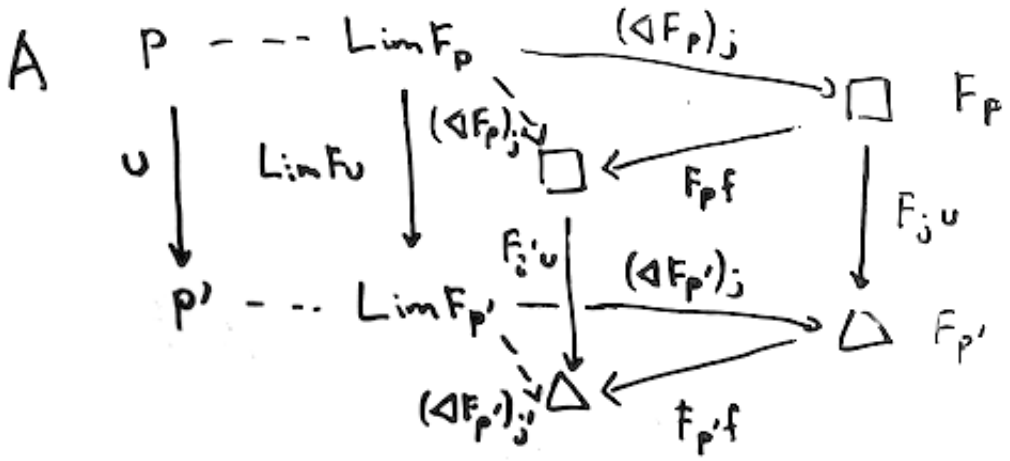
Limit
Lim F = LimF where F: J → A^P
LimF ∈ A^P

Limit cone
◁F: LimF ◁F
◁Fj: L ⇒ Fj

Corollary: continuity

Lim Fp = LimF p = (Lim F)p





==========

Proof

Idea: same as limit functor

- limit:

Pick any σ: T ⇒ F. Then for each p, there exists σ'p: Tp → Lim Fp such that

$\sigma p = \triangleleft Fp \circ \sigma' p$ by limit

claim: σ': T ⇒ L is well-defined

In particular, need to verify naturality.

****

claim proof

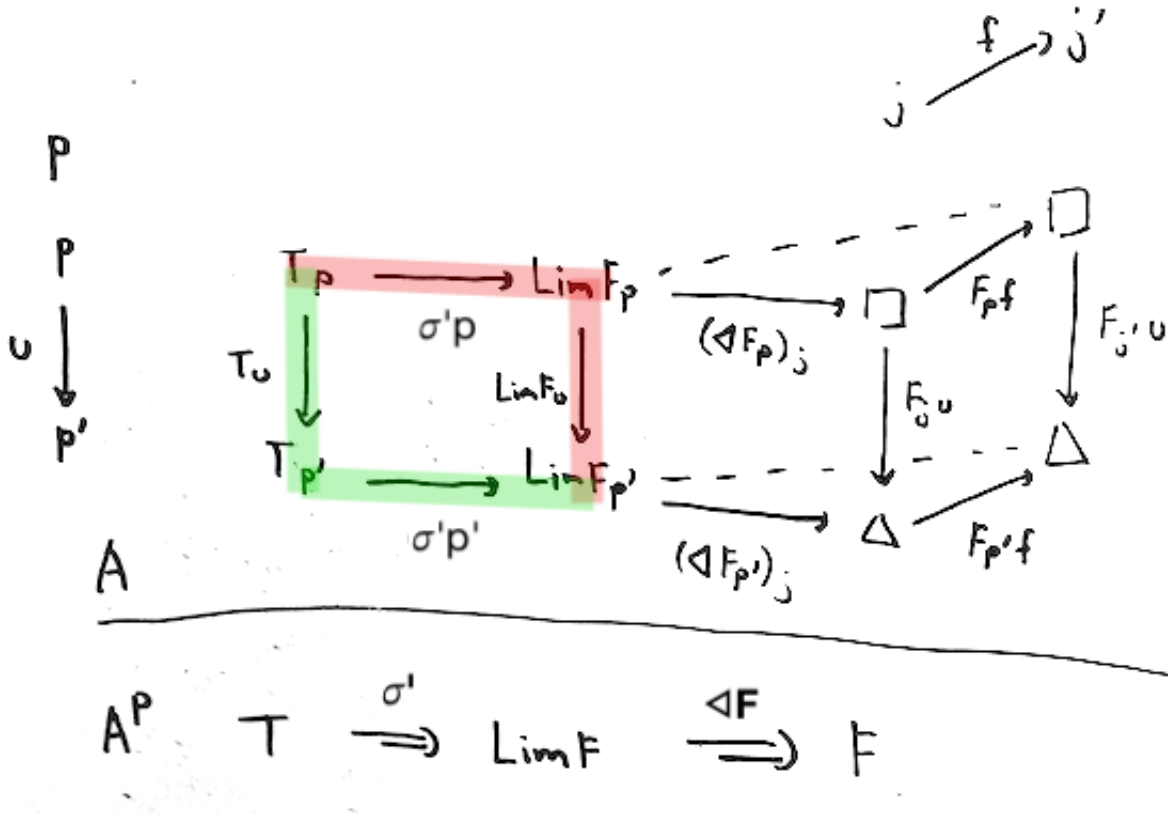
Idea: use uniqueness of limits

If u: p → p', then consider the cone Tp ◁ Fp'

$(\triangleleft Fp')j \circ (\sigma' p' \circ Tu) \quad = \quad ((\triangleleft Fp')j \circ \sigma' p') \circ Tu$

	=	σp' ◦ TU	by definition of σ
	=	Fju ◦ σp	
	=	Fju ◦ ((◁Fp)j ◦ σ'p)	
	=	(◁Fp')j ◦ (LimF u ◦ σ'p)	by (◁Fp')j ◦ LimF u = Fju ◦ (◁Fp)j by definition of LimF

=> (σ'p' ◦ Tu), (LimF u ◦ σ'p): Tp → Lim Fp both generate Tp ◁ Fp'



=> σ'p' ◦ Tu = LimF u ◦ σ'p by uniqueness of limits
=> σ': T ⇒ L is valid

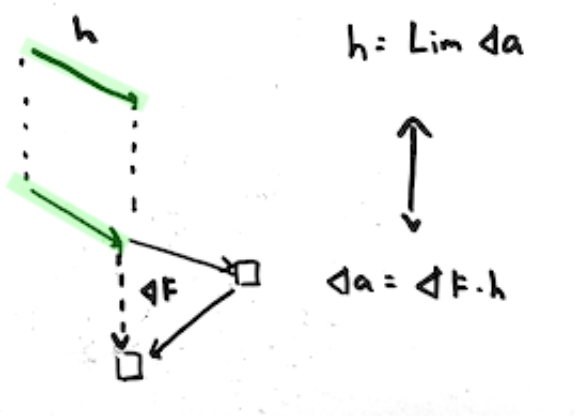
QED claim
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Hence σ = ◁F ◦ σ' by definition

QED
==========

Proposition: limit adjoint
If Lim F exists for every F:J→A (i.e. Lim is a functor), then *◁F ∈ [_→Lim F]◁F is an adjoint
That is, there is a natural 1-1 correspondence

Arrow		**Cone**
{ X → Lim F }	<=>	{ X ◁ F }
h: X →Lim F		◁a: X◁F
h = Lim ◁a		◁a = ◁F ◦ h



with

base map

◁F* ∈ [_→Lim F] ◁ F

◁F* (h) = ◁F ∘ h

co-base (inverse) map

Lim* ∈ Lim [X ◁ _] with limit functor Lim: A^J → A

Note: Lim X = X

Lim* (◁F ∘ h) = Lim (◁F ∘ h)

= h

Identity

◁F* ∘ Lim* (◁a) = ◁F * Lim (◁a)

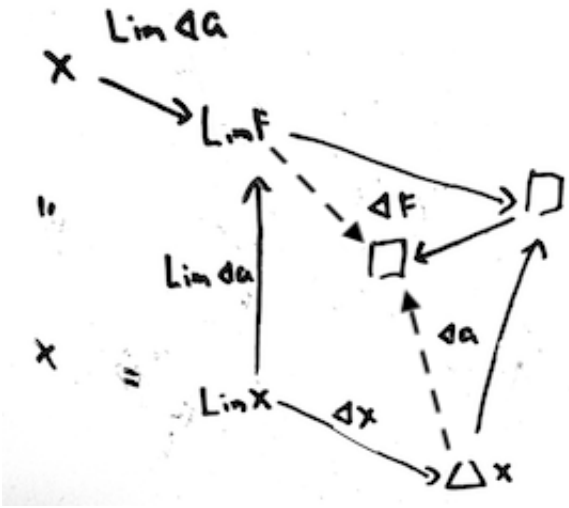
= ◁F ∘ Lim (◁a)

= ◁a

Lim* ∘ ◁F* (h) = Lim * ◁F (h)

= Lim (◁F ∘ h)

= h



Naturality

◁F'* ∘ *Lim σ = σ* ∘ ◁F* σ: F ⇒ F'

◁F' ∘ Lim σ = σ ∘ ◁F

with lattice

Base cone

◁: Lim ◁ I
◁F: Lim F ◁ F via whiskering ◁ * 1F = ◁F

Base co-cone
I: I ⇒ Lim **Δ** Identity
= I

Base identity
◁Lim ◦ Lim(I)
◁Lim ◦ Lim (F) = Lim (◁ (**Δ**Lim F))
=Lim F

Δ◁ ◦ I**Δ** = Identity
Δ◁ ◦ I**Δ** (X) = **Δ**(◁ **Δ**X)
=**Δ**X

==========

==== Preservation (aka continuity) ====

Limit preservation aka continuous
Limit creation
A functor F: A → B is **continuous** if it **preserves** base cones**.**
i.e. F is continuous if for any functor K: J → A

Lim K in A		**Lim FK** in B
◁K: Lim K◁K	=>	◁FK: Lim FK◁FK where
		Lim FK = F(Lim K)
		◁FK = F(◁K)

-
Dually, a functor F: A → B **creates limits** for A if it preserves cones in the opposite direction.
i.e. F creates limits for A if for any functor K: J → A

Lim FK in B		**Lim K** in A
◁FK Lim FK◁FK	=>	◁K: Lim K◁K where
		Lim FK = F(Lim K)
		◁FK = F(◁K)

e.g. the homset **{X→_}: A → Set** preserves limits (theorem below)

e.g. the forgetful **U: Grp → Set** creates limits (theorem below)

Proposition: composition
Let F: A → B, G: B → C
If F and G are continuous, then GF: A → C is continuous in C.
If F and G create limits, the GF: A → C creates limits for A.

Proposition: duality
If F: A → B creates limits and if a functor has at least one limit, then F also preserves limits for that functor.

i.e. If K: J → A and FK: J → B has a limit, then F preserves limits for K.

==========
proof:
Let ◁a: a◁K be a base cone of K created from FK i.e. F(◁a) is a base cone.
Then for any other base cone ◁a': a'◁K of K, there is an isomorphism f: a → a' such that

◁a' ◦ f = ◁a

=> F(◁a') ◦ Ff = F(◁a)
=> F(◁a') is isomorphic via Ff to a base cone F(◁a)
=> F(◁a') is a base cone.

QED
==========

Theorem: hom-functor continuity
Homset functors are coninuous
i.e. for a fixed X, let

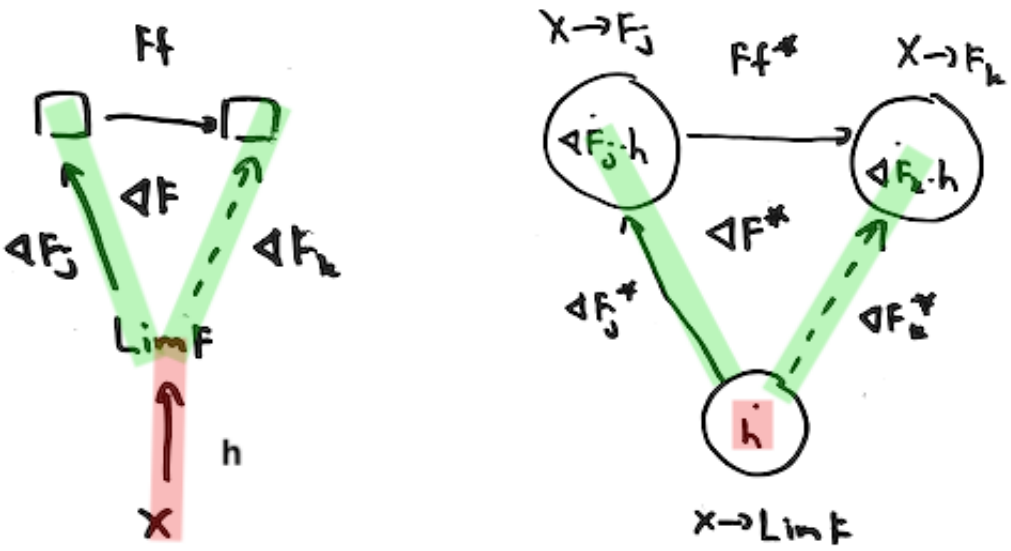
Hom-set functor
{ X → _ }: A → Set
f*: {X→Y} ⇒ {X→Y'} for any f:Y→Y'

Then for any functor F: J → A with a limit, the induced functor F*: J → Set has limit

Limit
Lim F* = {X→Lim F}

Cone
◁F*: Lim F* ◁F*
◁F*j: {X→Lim F} → {X→Fj}
◁F*j = (◁Fj)*

A **=>** **Induced**



Note: the induced functor F* is

F*: J → Set
F*j = {X → Fj} object map for j ∈ J

Ff*: {X→ Fj} → {X→Fk} arrow map for f:j→k
Ff(h) = Ff ◦ h

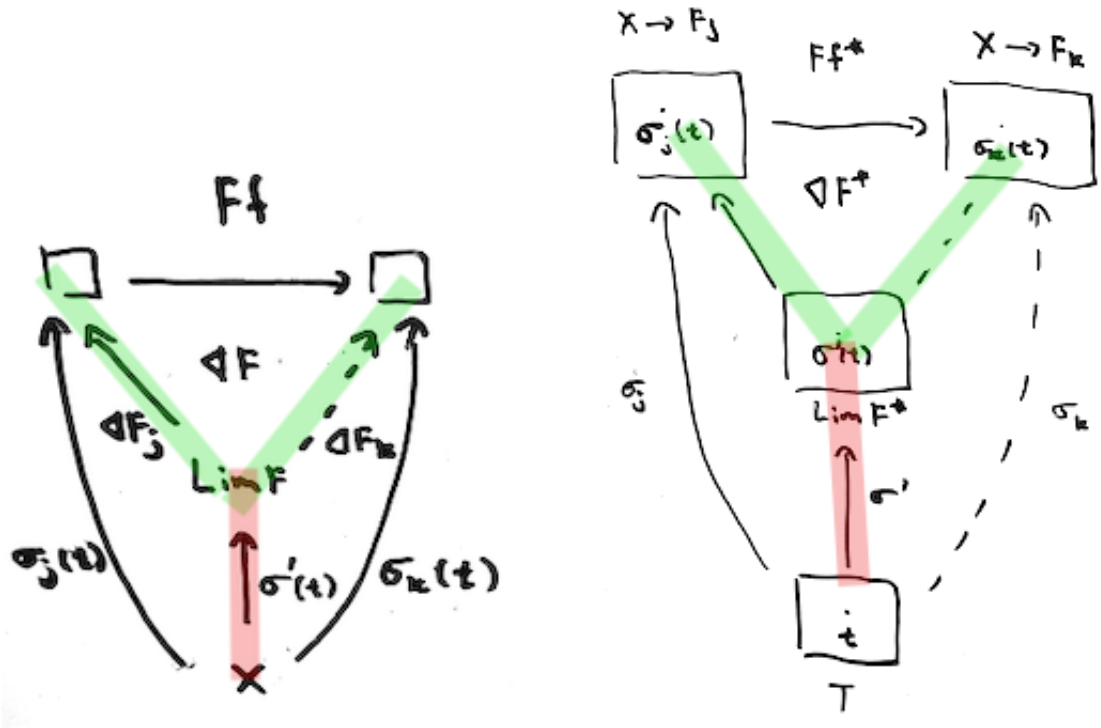
==========

Proof
Let any σ: T◁F*.
Then for any t ∈ T

σj(t): X → Fj
σk(t) = Ff ◦ σj(t) for any f:j→k

=> σ(t) is a cone X◁F
=> there exists σ'(t): X→Lim F such that
σ(t) = ◁F ◦ σ'(t)
=> σ = ◁F* ◦ σ'

A => **Homset**



QED claim
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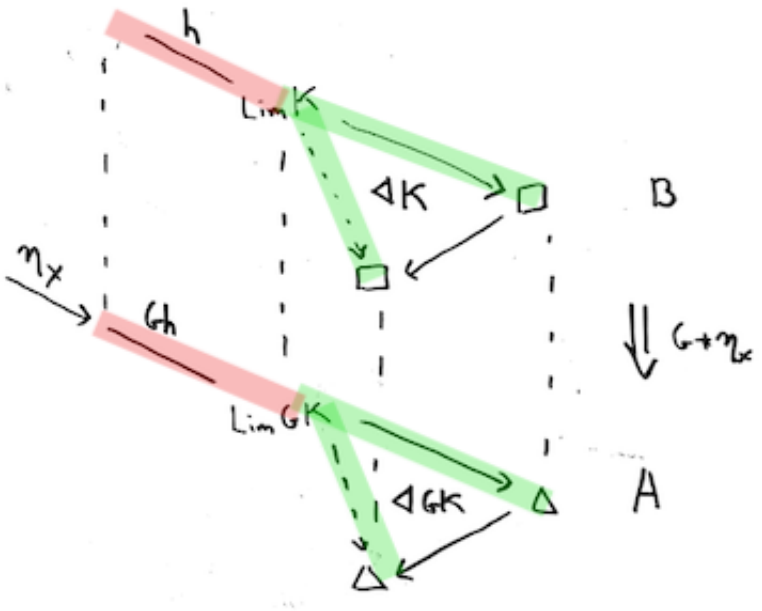
Proposition: adjoint continuity
Adjoints are continuous
i.e. Let
F: A→B G:B→A
K: J →B
G * η ∈ ∗→G[F∗→_]

If G * η is an adjoint and Lim K exists, then Lim GK exists where

Limit	**Base cone**
Lim GK = G(Lim K)	◁(GK) = G(◁K)

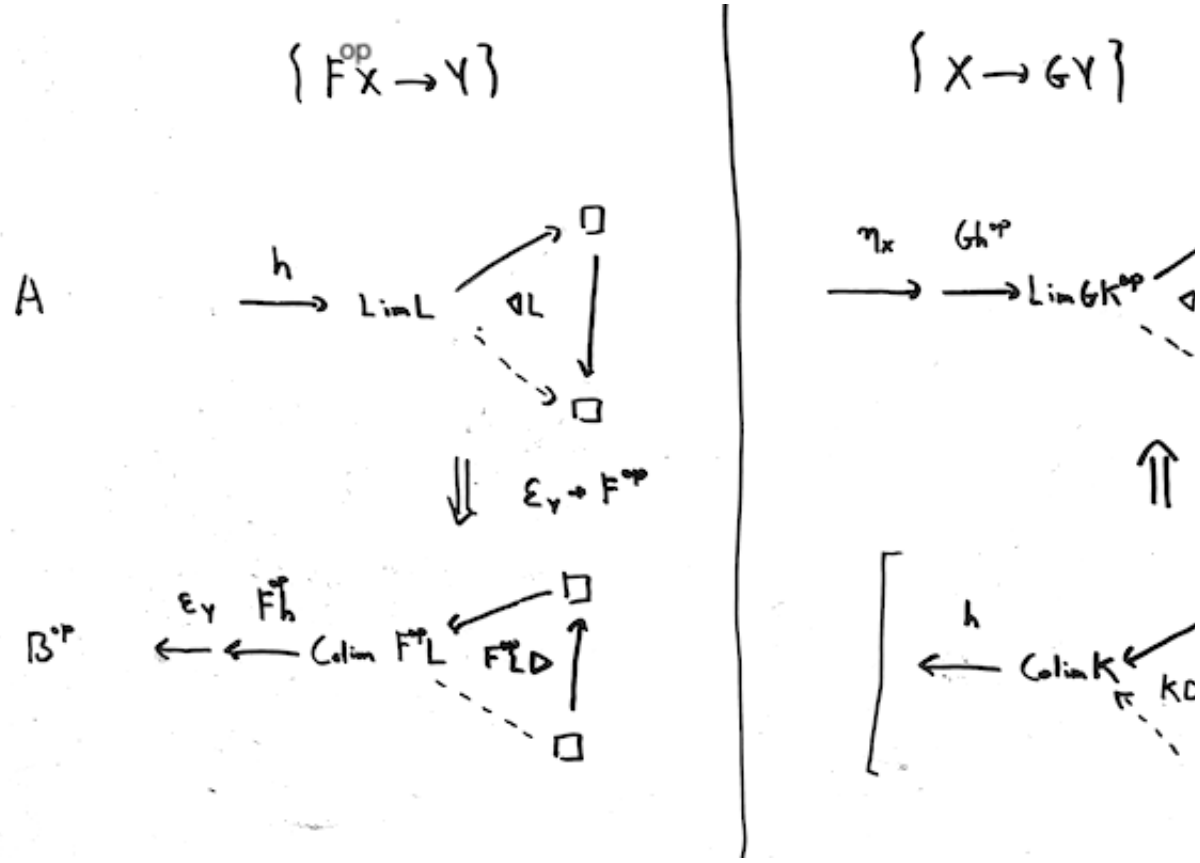
To be precise:

	◁(GK)* ∘ (G * η)	**=**	**(G * η) ∘ (◁K)***
base map	◁(GK) ◦ (Gh ◦ ηX)	=	G[◁K ◦ h] ◦ ηX
Offset	Gh ◦ ηX: X → Lim GK	<=>	h: FX → Lim K



Corollary: opposites
Let functors
K: J → B
L: J → A

	B^op	**A**
Adjoint	{ F^opX → Y} F^op: A → B^op	{ X → GY } G: B^op → A
Limit	**limit of A** as colimit in B^op Colim F^opL = F^op(Lim L)	**colimit of B** as limit in A Lim GK^op = G(Lim K^op) = G((Colim K)^op)
Cone	(F^opL)▷ = F^op(◁L)	◁(GK^op) = G(◁K^op) = G((K▷)^op)



	B	**A^op**
Adjoint	{ FX → Y } F: A^op → B	{ X → G^opY } G^op: B → A^op
Limit	**limit of A** as colimit in B Colim FL^op = F(Colim L^op) = F((Lim L)^op)	**colimit of B** as limits in A^op Lim G^opK = G^op(Colim K)
cone	(FL^op)▷ = F(▷L^op) = F((◁L)^op)	◁(G^opK) = G^op(K▷)

==========

Proof Theorem

Idea: just by **composition** of isomorphisms ◁K* and G * η

i.e.

$G(\lhd K)^* \circ (G * \eta) = (G * \eta) \circ \lhd K^*$

Pick any $\lhd\sigma$: X◁ GK

Then ◁K* and G * η are both isomorphic, so

$\lhd\sigma = (G * \eta X) \circ \lhd K^*(h)$ for some unique h: X→Lim K

$\Rightarrow \sigma = (G * \eta X) \circ \lhd K^*(h)$

$= G(\lhd K)^* \circ (G * \eta X)(h)$ by composition

=> h' = G * ηX (h) is the unique arrow such that σ = G(◁K) (h')

Hence ◁(GK) = G(◁K)

QED
==========

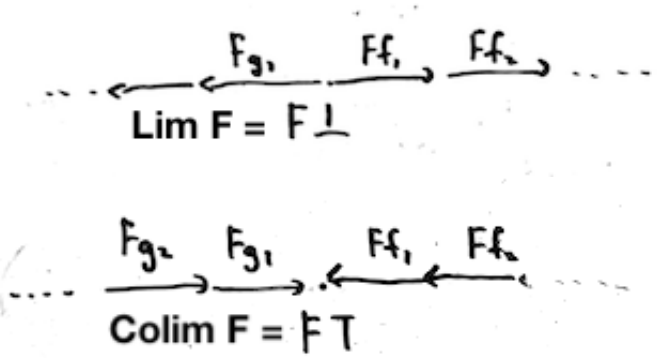
==========
==== Creation ====

Theorem: initial/terminal
Let F: J → A
If J has an initial object ⊤, then F has a limit

Lim F = F⊥
◁Fi: F⊤ → Fi

If J has an terminal object ⊥, then F has a colimit

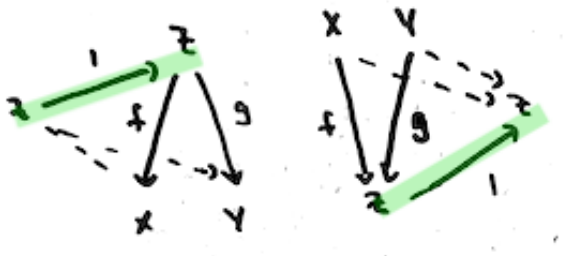
Colim F = F⊥
F▷i: Fi → F⊥



Corollary:

f: Z → X, g: Z → Y
Lim [f∧g] = Z
Colim [f∧g = X ⊔ Y

f: X → Z, g: Y → Z
Lim [f∨g] = X x Y
Colim [f∨g] = Z



==========
Proof:
- initial object ⊤
Suppose ◁a: X◁F. Let f = ◁a⊤: X → F⊤
Then for any j ∈ J, there is a unique

◁Fj: F⊤ → Fj

=> ◁aj = ◁Fj ◦ f

=> ◁a = ◁F ◦ f

QED
==========

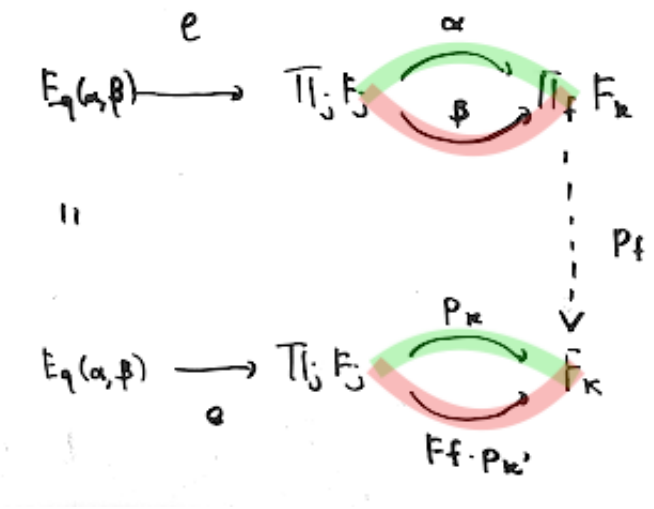
Theorem: limit
Let A be a category with all products and equalizers.
Then every functor has a limit.

To be precise, if F: J → A is a functor, consider the system of equations α, β: ∏j Fj → ∏f Fk for each f: k' → k

pf ◦ α = pk : ∏j Fj → Fk
pf ◦ β = Ff ◦ pk' : ∏j Fj → Fk

Equate them with the equalizer e: Eq(α, β) → ∏j Fj, so that

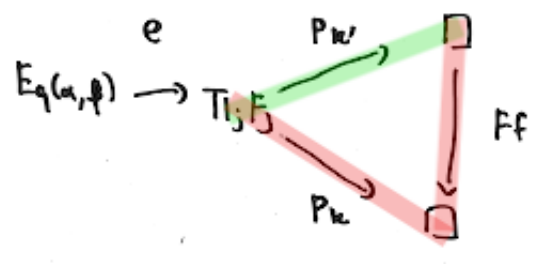
pk ◦ e = (pf ◦ α) ◦ e = (pf ◦ β) ◦ e = (Ff ◦ pk') ◦ e



Then the equalizer is the limit

Limit
Lim F = Eq (α, β)

Base
◁F: Eq (α, β) ◁ F
◁F = ∧j pj ◦ e where ◁Fj = pj ◦ e: Eq(α, β) → Fj for any j ∈ J



Theorem: colimit
If F: J → A a functor, consider the system of equations α, β: $\sqcup_f$ Fk → $\sqcup_j$ Fj for each f: k → k'

$\alpha \circ i_f = i_k$: Fk → $\sqcup_j$ Fj
$\beta \circ i_f = i_{k'} \circ Ff$: Fk → $\sqcup_j$ Fj

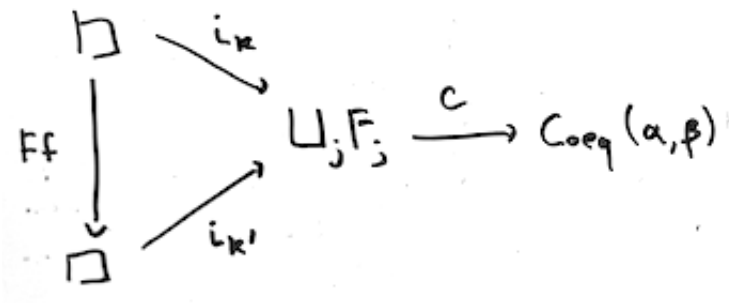
Equate them with the coequalizer c: $\sqcup_j$ Fj → Coeq(α, β) so

$c \circ i_k = c \circ (\alpha \circ i_f) = c \circ (\beta \circ i_f) = c \circ (i_{k'} \circ Ff)$

Then the coequalizer is the colimit

Collimate
Colim F = Coeq (α, β)

Cobase
F▷: F▷ Coeq (α, β)
F▷ = c ∘ ∨j ij where F▷j = c ∘ ij: Fj → Coeq(α, β) for any j ∈ J



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Proof:

claim: ∏j Fj ∘ eq is indeed a cone
i.e. satisfies Ff ∘ (pj ∘ eq) = (pj' ∘ eq) for each f: j' → j

proof claim:

pj ∘ eq = pf ∘ α ∘ e by definition of α
= pf ∘ β ∘ e by equalizer
= (Ff ∘ pj') ∘ e by β
= Ff ∘ (pj' ∘ e)
QED claim

claim: ∧j pj ∘ eq is a base

proof claim: Let any ◁a: Z◁F
Let a': Z → ∏j Fj be the corresponding product form i.e.

◁a = ∧j pj ∘ a'

Then for any f: j' → j

pf ∘ α ∘ a' = pj ∘ a'
= Ff ∘ pj' ∘ a' by ◁a
= pf ∘ β ∘ a'

=> a' equalizes (α, β)
=> a' = eq ◦ h unique h: Z → Lim F
=> ◁a = ∧j pj ◦ (eq ◦ h)
= (∧j pj ◦ eq) ◦ h

=> ∧j pj ◦ eq is a base ◁F

QED claim
==========

Corollary: complete sets

- **Set** is complete
For any functor F: J → Set, the limit is just the set of all combinations aka product that satisfy the triangle relation

Lim F = { x ∈ ∏j Fj: Ff(xj) = xj' for any f: j → j' }

◁Fj (x) = xj jth projection

- Forgetful functor **U: Grp → Set** creates limits
Hence Grp is complete

Likewise for Vect, Mod, etc

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Proof

- **Set**: Set has products and equalizers
By the construction, Lim F is the equalizer of

α(x ∈ ∏j Fj) = ∏f Ff(xj)

β(x ∈ ∏j Fj) = ∏f xj' for f: j → j'

=> equalize

Lim F = Eq(α, β) = { x ∈ ∏f Fj: Ff(xj) = xj' for any f: j → j' }

- U: Grp → Set
Just need to verify that
a) Lim F can be made into a **group**
b) each ◁Fj: Lim F → Fj can be made into a **homomorphism**

a) Let

product xy = ∏j (xj · yj)
inverse x^-1 = ∏j xj^-1

These are valid since

Ff(xy)j = Ff(xj)Ff(yj) Ff homomorphism
= xj'yj'
= (xy)j'

Ff(x^-1j) = Ff(xj)-1 Ff homomorphism
= xj^-1
= (x^-1)j

b) projection ◁Fj is trivially homomorphic.

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==== Product/coproduct ====

Theorem: expressions
A product/coproduct is a way to represent multiple expressions as single.
If A has any products

fn: X → Yn for each n <=> f: X → ∏n Yn
fn = pn ◦ f

If A has any coproducts

gn: Xn → Y for each n <=> g: ⊔n Xn → Y
gn = g ◦ in

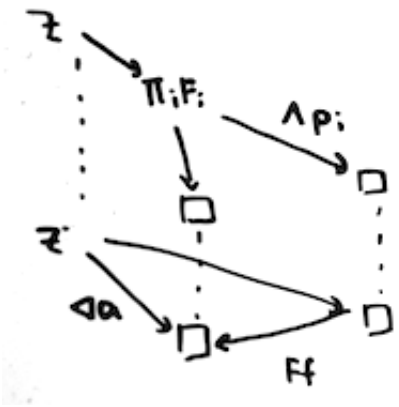
Theorem: product form
A cone can be expressed as a product
i.e. Let F: J → A.
If A has all products, then

◁a: Z◁F cone
a': Z → ∏n Fn product form

◁a = ∧n pn ◦ a' projection pm: ∏n Fn → Fm
◁an = pn ◦ a'

note: NOT every product is a cone since it needs to satisfy the triangle

pn ◦ a' = Ff ◦ (pn ◦ a')



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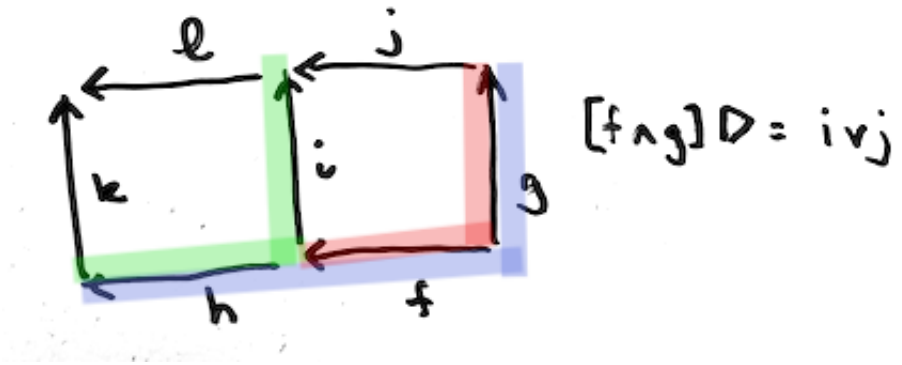
==== Pushout/pullback ====

Proposition: pushout

Let squares

$f \wedge g$ with colimit $[f \wedge g]\triangleright = i \vee j$

$h \wedge i$

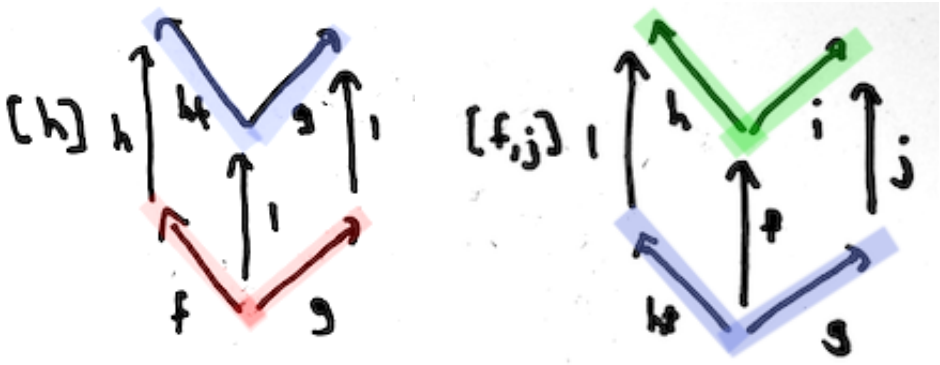


Related by natural transforms

$[h]: [f \wedge g] \Rightarrow [hf \wedge g]$ where $[h]_codom(f) = h$

$[f, j]: [hf \wedge g] \Rightarrow [h \wedge i]$ where $[f, j]_dom(g) = f$

$[f, j]_codom(g) = j$



a) If $[h \wedge i]$ has a colimit with base $[h \wedge i]\triangleright$ $k \vee l$, then $[hf \wedge g]$ also has a colimit.

Object

Colim $[hf \wedge g]$ = Colim $[h \wedge i]$

$[hf \wedge g]\triangleright$ = $k \vee lj$

Transformation

Colim $[h]$ = l

$[hf \wedge g]\triangleright \circ [h]$ = Colim $[h] \circ [f \wedge g]\triangleright$

$k \vee lj \circ [h]$ = $kh \vee lj$

= $li \vee lj$

= $l \circ i \vee j$

b) If $[hf \wedge g]$ has a colimit with base $[hf \wedge g]\triangleright = k \vee m$, then $[h \wedge i]$ has a colimit

Object

Colim [h ∧ i] = Colim [hf ∧ g]

[h ∧ i]▷ = k ∨ l where l is the unique arrow kh ∨ m: [f ∧ g]▷ codom m

kh ∨ m = l ∘ [f ∧ g]▷

i.e. **m = lj**

Transformation

Colim [f, j] = 1

[h ∧ i]▷ ∘ [f, j] = Colim [f, j] ∘ [hf ∧ g]▷

= [hf ∧ g]▷

k ∨ l ∘ [f,j] = k ∨ lj

= k ∨ m

==========

proof:

a) Let any cone a▷ = s ∨ t: [hf ∧ g] ▷ X. Then

sh ∨ t: [f ∧ g]▷X

=> sh ∨ t = r ∘ [f ∧ g]▷ for some unique r

= ri ∨ rj

=> sh = ri t = rj

Then also

s ∨ r: [h ∧ i]▷X

=> s ∨ r = r' ∘ [h ∧ i]▷X for some unique r'

= r'k ∨ r'l

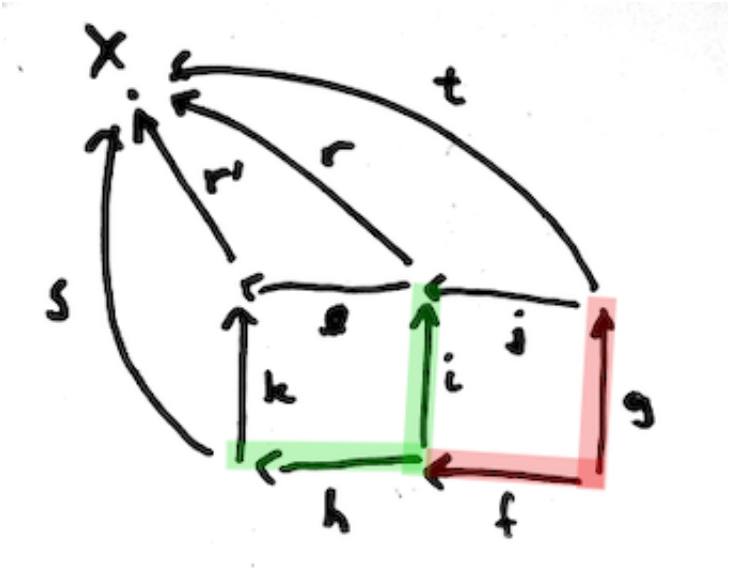
=> s = r'k

t = rj = r'lj

=> a▷ = s ∨ t

= r' ∘ k ∨ lj

Hence [hf ∧ g]▷ = k ∨ lj



QED

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==== OLD ====

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is
code
Monaco 14
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